

MSDM5004 Spring 2026

Project 1 Part I

Due Apr. 12

You are required to submit a report and all the codes for a project. Don't save the result of every time step in the evolution.

1. Consider the problem

$$\begin{cases} \frac{\partial u}{\partial t} = a \frac{\partial^2 u}{\partial x^2} & -1 < x < 1, \quad t > 0 \\ u(-1, t) = u(1, t) = 0 & t > 0 \\ u(x, 0) = u_0(x) & -1 \leq x \leq 1 \end{cases}$$

where $a = 2$ and

$$u_0(x) = \begin{cases} \frac{1}{3}(x+1) & -1 \leq x \leq \frac{1}{2} \\ 1-x & \frac{1}{2} < x \leq 1. \end{cases}$$

(1) Obtain numerical solution using the explicit scheme

$$U_j^{n+1} = U_j^n + a\mu(U_{j+1}^n - 2U_j^n + U_{j-1}^n),$$

where $\mu = \frac{\Delta t}{(\Delta x)^2}$. Use $J = 20$, $\Delta x = 0.1$, and (i) $\Delta t = 0.0025$ (ii) $\Delta t = 0.0026$. Plot the numerical solution at $t = 0, \Delta t, 25\Delta t, 50\Delta t$.

(2) Obtain numerical solution using the Crank-Nicolson method. You are required to use the Thomas algorithm to solve the linear system in the numerical discretization.

Use $J = 20$, $\Delta x = 0.1$, and (i) $\Delta t = 0.0025$ (ii) $\Delta t = 0.0026$. Plot the numerical solution at $t = 0, \Delta t, 25\Delta t, 50\Delta t$.